



22CH203 - ENGINEERING CHEMISTRY II

UNIT III & LP 3 – METAL CORROSION AND ITS PREVENTION

1. CORROSION CONTROL

The rate of corrosion can be controlled by either **modifying the metal or the environment.**

1.1. Control of corrosion by modifying the metal

1.1.1. By selection of the metal:

The **selection of the right type of metal** is the main factor for corrosion control. Thus, noble metals are used in ornaments and surgical instruments, as they are most immune to corrosion.

1.1.2. By using pure metal:

Pure metals have higher corrosion resistance. Even a small amount of impurity may lead to severe corrosion.

1.1.3. By alloying:

The corrosion resistance of many metals can be improved **by alloying.** For example, stainless steel containing chromium produces a coherent oxide film, which protects the steel from further attack.

1.1.4. By proper design

Some of the important rules for designing, which must be observed are given below.

(i) Avoid galvanic corrosion

If two different metals are joined, galvanic corrosion will occur. In such a case galvanic corrosion is prevented by

- Selecting the metals as closely as possible in the electrochemical series.**
- Providing a smaller area for the cathode and a larger area for the anode.**
- Inserting an insulating material between the two metals**

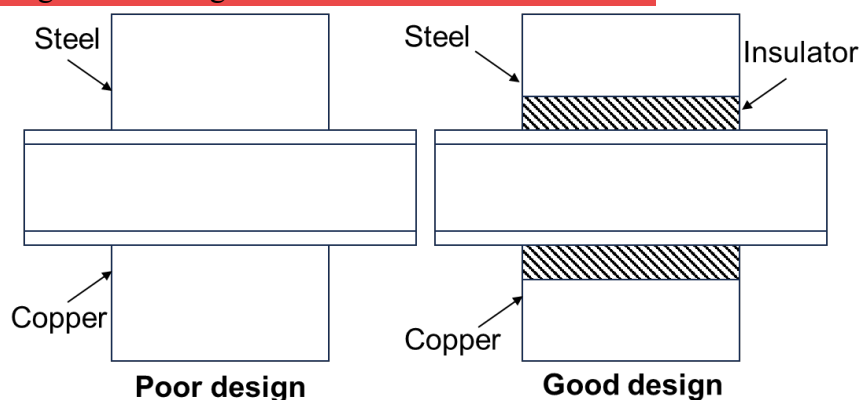


Fig. 3.1. Insulation avoids galvanic corrosion

(ii) Drainage affects corrosion:

Tanks and other containers must be designed in such a way that the whole of the **liquid** should be **drained off completely.**

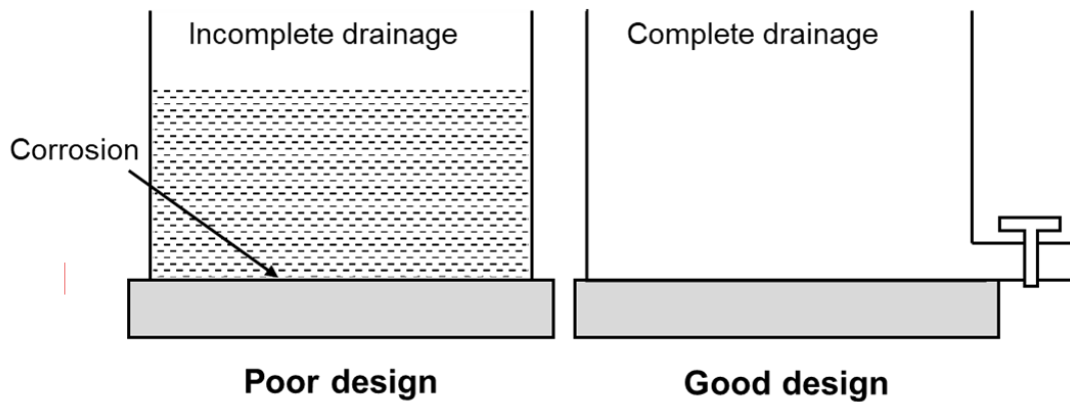


Fig. 3.2. Drainage affects corrosion

(iii) Avoid sharp corners and bends:

Sharp corners or edges should always be avoided, and hence erosion corrosion can be avoided by **smooth corners or curved pipe bends**.

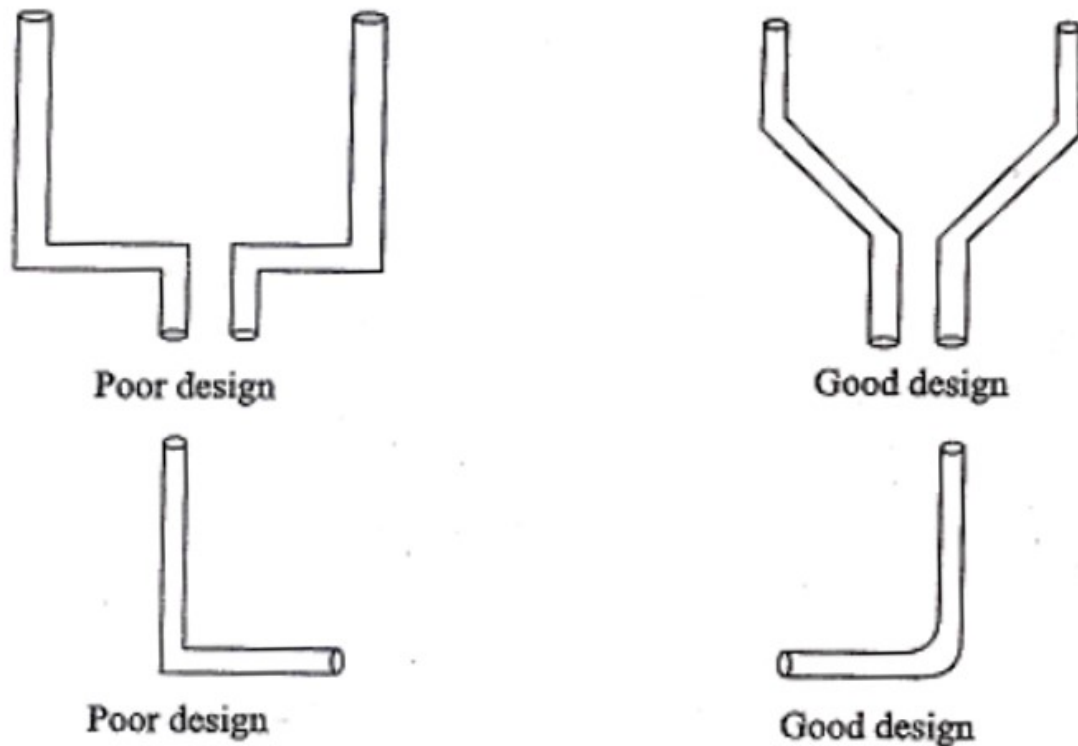


Fig. 3.3. Corrosion prevention using curved pipeline

(iv) Avoid crevices:

Crevice corrosion is a localized form of corrosion that occurs in the crevices of a metal structure. It is caused by the accumulation of moisture and dirt, which results in increased electrochemical corrosion. This can be prevented by filling the crevices with a filler.

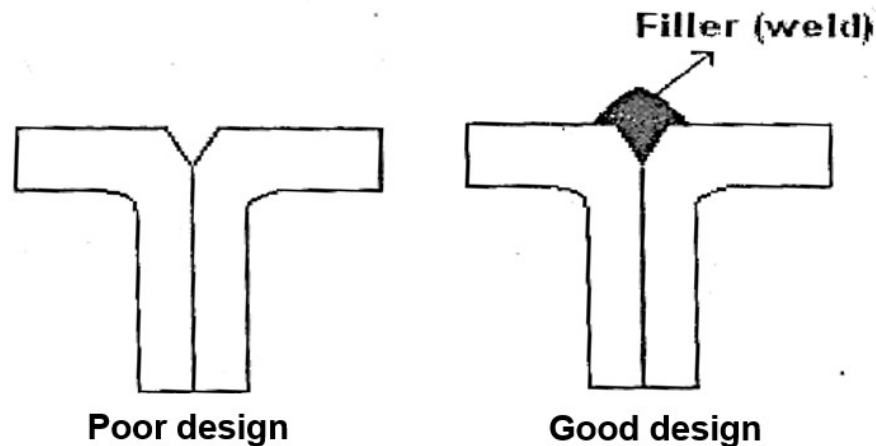


Fig. 3.4. Minimizing crevice corrosion

1.2. Electrochemical Protection

The principle involved in electrochemical protection (or) cathodic protection is to **force the metal to behave like a cathode**. The important cathodic protections are

- Sacrificial anodic protection
- Impressed current cathodic protection

1.2.1. Sacrificial anodic protection method

In this method, the metallic structure to be protected is made a cathode by connecting it with **more active metal** (anodic Fig. 3.5 metal). The principle involved in electrochemical protection (or) cathodic protection is to force the metal to behave like a cathode. The corrosion will concentrate only on the active metal. The artificially made anode thus gradually gets corroded protecting the original metallic structure. Hence this process is otherwise known as sacrificial anodic protection. Aluminium and Zinc are used as sacrificial anodes.

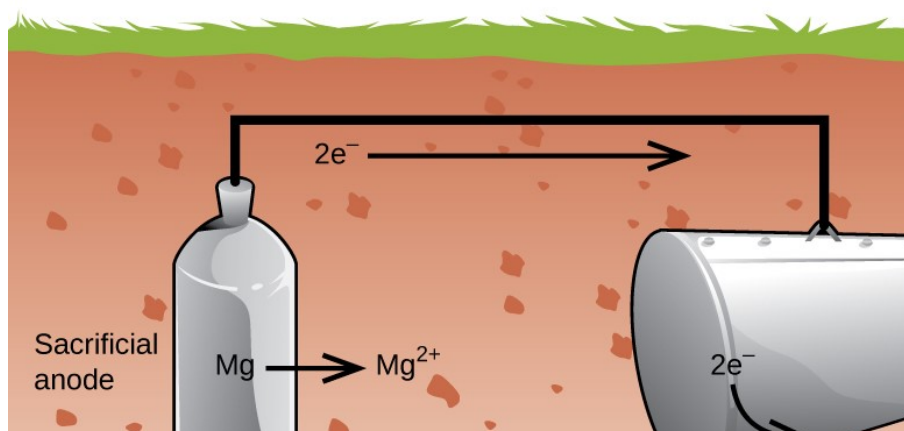


Fig. 3.5 Sacrificial anodic protection.

1.2.2. Applications of sacrificial anodic protection

- This method is used for the protection of ships and boats (fig 3.6). Sheets of Mg or Zn are hung around the hull of the ship Zn or Mg will act as an anode compared to iron (the ship or

boat is made of iron), so corrosion concentrates on Zn or Mg. Since they are sacrificed in the process of saving iron, they are called sacrificial anodes.

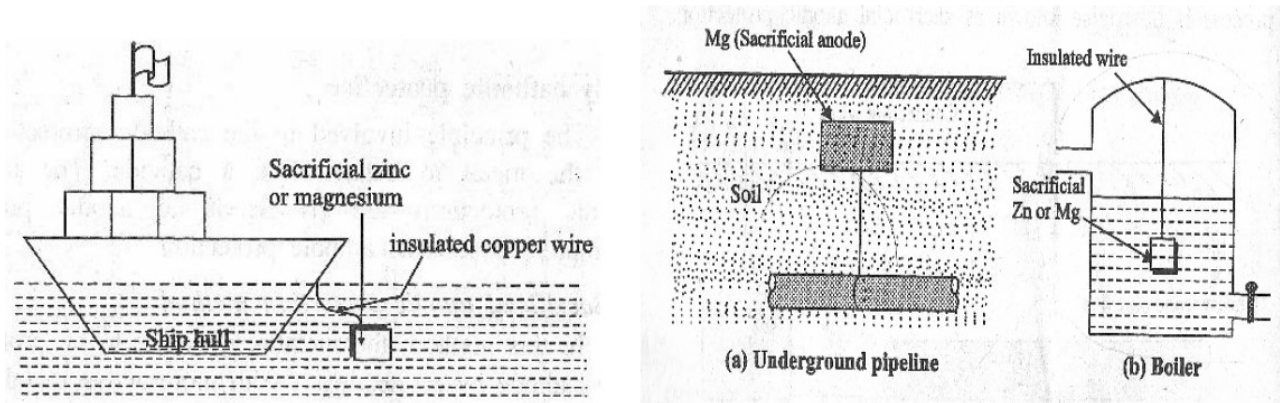


Fig. 3.6. Applications of sacrificial anodic protection.

- (ii) Protection of underground pipelines, and cables from soil corrosion (Fig. 3.6(a)).
- (iii) Insertion of Mg sheets into the domestic water boilers to prevent the formation of rust (Fig. 3.6(b)).
- (iv) Calcium metal is employed to minimize engine corrosion.

1.2.3. Impressed current cathodic protection method

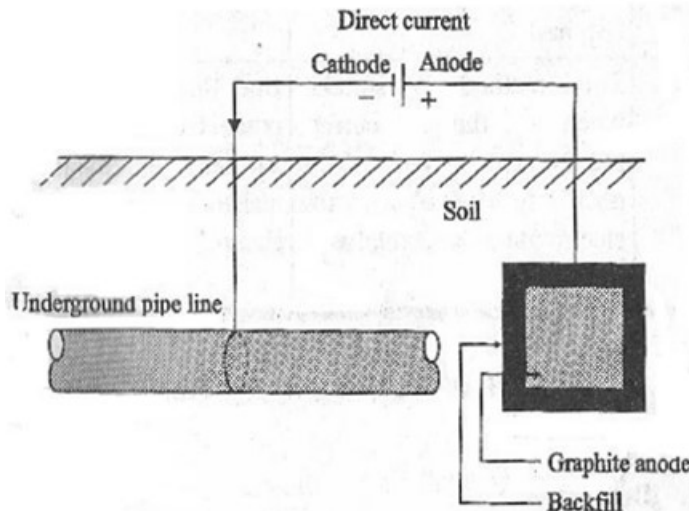


Fig. 3.7. Impressed current cathodic protection.

In this method, an impressed current is applied in the opposite direction of the corrosion current to nullify it, and the corroding metal is converted from anode to cathode (Fig.3.7).

This can be done by connecting the negative terminal of the battery to the metallic structure, to be protected, and the positive terminal of the battery is connected to an inert anode. Inert anodes used for this purpose are graphite and platinized titanium. The anode is buried in a “backfill” (containing a mixture of gypsum, coke, breeze, and sodium sulphate). The “backfill” provides good electrical contact to the anode.

1.2.4. Applications of impressed current protection

Structures like tanks, pipelines, transmission line towers, underground water pipelines, oil pipelines, ships, etc., can be protected by this method.

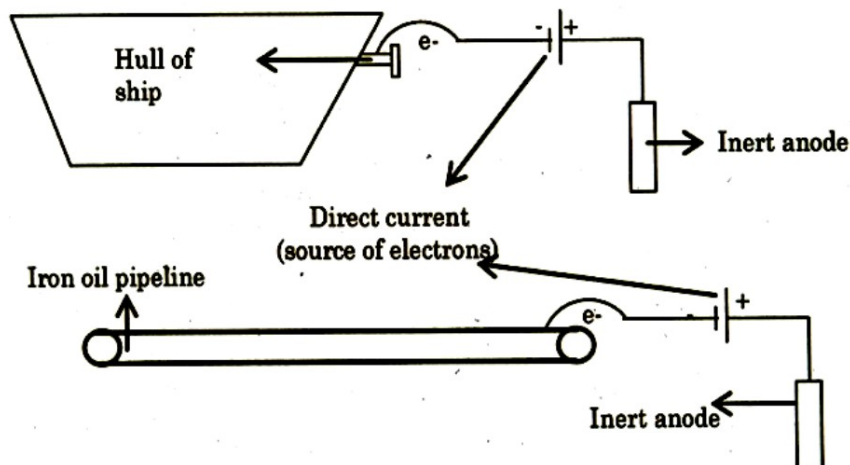


Fig. 3.8. Applications of impressed current cathodic protection.

1.2.5. Difference Between Sacrificial Anode and Impressed Current Method

S. No	SACRIFICIAL ANODE METHOD	IMPRESSED CATHODIC CURRENT PROTECTION METHOD
1	No external power supply is necessary.	An external power supply must be present.
2	This method requires the periodical replacement of the sacrificial anode.	Here anodes are stable and do not disintegrate.
3	Investment is low.	Investment is high.
4	Soil and microbiological corrosion effects are not taken into account.	Soil and microbiological effects are taken into account.
5	This is the most economical method especially when short-term protection is required.	This method is well-suited for large structures and long-term operations.
6	This method is suitable when the current requirement and the resistivity of the electrolytes are relatively low.	This method can be practiced even if the current requirement and the resistivity of the electrolytes are high.



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited

SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU

Discussion questions:

1. Apply your knowledge of corrosion factors to develop corrosion prevention strategies for industry.
2. Explain the relationship between the purity of steel and its susceptibility to corrosion.
3. Propose two specific design modifications you can implement to minimize the corrosion issue.
4. What materials and techniques are best suited to prevent corrosion on a ship's hull?